Categories: Implicit and Imperfect

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Analysis, Evaluation, and Commentary Regarding Squire and Knowlton (1995)

Brief Summary of Squire and Knowlton (1995)

Squire and Knowlton (1995) tackle the question of how people without the benefit of a permanent long-term memory mentally categorize knowledge in their minds. The study uses the case of an amnesiac patient, E.P, whose memory is too impaired to have any declarative (explicit) memory. E.P. and four healthy control subjects were measured across four different tests on their ability to categorize or recognise patterns based on previous 'training' stimuli. The training stimuli were either distortions of a prototype – the most typical exemplar of a category – or the protype repeated in succession. In *Classification* stages, subjects were asked to classify stimuli based on training stimuli; in *Recognition* stages, they were merely asked whether or not they had seen the stimulus previously. The results showed that E.P. performed significantly better in the former case, but not the latter case. We can extrapolate, therefore, that functional declarative memory that has been proven necessary for an individual stimulus is not necessary for categorization – instead, this is an implicit process, potentially using a separate memory system.

Analysis of Squire and Knowlton (1995)

Patient E.P. was chosen as a candidate with severe anterograde and retrograde amnesia, meaning that he could neither remember recent memories nor create new memories. The four control volunteers were closely matched with E.P. on other criteria, including age and total number of years of education – resulting in a restricted participant pool. This also restricts the generalizability of the results: as demonstrated by Verheyen et al. (2019), there are significant differences between young adults and older adults in semantic categorization tasks – particularly relevant is that older adults tended to have larger "category extensions" (p. 12) and were more permissive in accepting items to categories. In this view, it is likely that, if the experiment is replicated, healthy younger participants may not match younger

amnesiacs on the classification tasks. Thus, the study would benefit from a broader participant pool.

The results showed a dissociation between recognition and classification abilities of E.P. – leading the Squire and Knowlton (1995) to theorize that there must be two "separate and parallel memory function" (p. 12470) for these two processes. However, Nosofsky and Zaki (1998) contended this conclusion by comparing the performance of patients with Parkinson's disease with good memory on classification and recognition tasks. The PD patients had significantly worse early stages of classification but performed at par with E.P. and healthy participants in late classification – showing that classification cannot be attributed to one system of memory itself, and shedding doubt on the theory of a dual system.

Implications of Squire and Knowlton (1995)

Squire and Knowlton (1995) concluded that since E.P, who has no declarative memory, had the ability to classify, the process of classification is non-declarative. Recognition, on the other hand, is purely declarative, as evidenced by E.P's significantly worse performance on recognition tests. The process of categorization being non-declarative is in some ways unsurprising – categorization is necessary to handle the constant stimuli of the world and eliminate the need for constant learning. However, the fact that people can learn unconsciously is especially useful in training people with lowered cognitive function – amnesiacs such as E.P, children born with brain damage or other disabilities. However, the applications of this ability to categorize are limited – it is unclear whether the amnesiac patients can actually name the category they have learned or be able to manipulate the objects after being able to identify them.

Applying the Psychology of Categories to Everyday Life

The Situation: Us!

Memorizing information is vital to many fields – and undoubtedly, students make up much of the global population in this regard. Though implicit categorization aids us on a

sensory level, this ability does not automatically occur for semantic learning. However, the findings from Squire and Knowlton (1995) show that it *is* possible to learn new information with less effort using categorization processes.

Given training, students can categorize information by themselves. For example, upon encountering an academic paper where the participants are asked whether they have encountered a stimulus before, students can file it as a 'recognition task'. Based on what the participants are asked to recognize, the student can further infer whether the paper discusses a model of attenuation or LTM constructs. Moreover, instructors can structure their courses in a way that applies these categorical rules – helping students semantically organize recognition tasks across a range of topics. This may help student learn details more easily, considering they already know the superordinate category (Reed, 2013, p. 184).

Relevant Psychological Theories and/or Principles

Categorization depends on the population of a category rather than the category itself. Especially relevant is typicality – a measure of how well a single category member represents that category. Typicality arises from family resemblance – a measure of how frequently attributes of a category are shared among members. Some members may have fewer attributes and thus represent their category worse than other members, leading to miscategorization (or slower and less efficient categorization) of new stimuli that are similarly atypical.

Interpretation According to the Psychology of Categories

Prototypes are the most typical and have most family resemblance of their category; however, while they are useful for dot patterns which occur along a continuous dimension, it may pose problems for recognition tasks which vary across a continuum. Thus, implicit categorization, which is unconscious, suffers doubly – by typicality *and* unconscious errors that cannot be actively corrected.

References

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